PORTFOLIO ARTIFACT · AIML-501 MODEL DEVELOPMENT

Machine Learning Algorithm Framework

Eleven core algorithms grouped by how they learn, mapped to the domains they are used in, from classic tabular models to the deep learning behind today's generative AI.

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DOMAINS · Tabular Data · Computer Vision · NLP · Generative AI

Supervised
Learns from labeled examples to predict a known answer

01 Linear & Logistic Regression
Tabular

Fits a weighted line (or S-curve for logistic) through the data. Each feature gets a weight showing how much it pushes the prediction up or down.

Use cases: house price prediction, credit default risk, email spam yes/no.

02 Decision Tree
Tabular

Splits data with a series of yes/no questions (e.g. "income > $50k?") until each branch ends in a prediction. Easy to read and explain.

Use cases: loan approval rules, medical triage, customer churn.

03 Random Forest & Gradient Boosting
Tabular

Combines many decision trees. Random Forest averages independent trees; Gradient Boosting (XGBoost, LightGBM) builds trees one after another, each fixing the last one's errors.

Use cases: fraud detection, demand forecasting, ranking. Often the top performer on spreadsheet-style data.

04 K-Nearest Neighbors (KNN)
Tabular · Vision

Labels a new point by looking at the K most similar labeled points and taking a majority vote. No real training step, it just compares.

Use cases: product recommendations, simple image matching, filling missing values.

05 Convolutional Neural Network (CNN)
Vision

Slides small filters across an image to detect edges, then shapes, then whole objects, layer by layer. Built to understand pixels.

Use cases: face ID, medical X-ray analysis, self-driving car perception, defect inspection.

06 Transformer
NLP · GenAI · Vision

Uses attention so every word can "look at" every other word to understand context, then predicts the next token. The architecture behind LLMs like ChatGPT and Claude.

Use cases: chatbots, translation, summarization, code assistants.

Pre-trained self-supervised on raw text (the text is its own label), then fine-tuned with supervision.

Unsupervised
Finds structure in data that has no labels

07 K-Means Clustering
Tabular

Places K center points, assigns each data point to the nearest center, moves the centers to the middle of their group, and repeats until the groups stop changing.

Use cases: customer segmentation, grouping documents, spotting unusual login behavior.

08 Principal Component Analysis (PCA)
Tabular · Vision

Compresses many features into a few new ones that keep most of the variation. Makes data smaller, faster to train on, and easier to plot.

Use cases: reducing hundreds of sensor readings, image compression, visualizing high-dimensional data.

09 Autoencoder
Vision · Tabular · GenAI

A neural network that squeezes input into a small code and then rebuilds it. If it can't rebuild something well, that input is probably unusual.

Use cases: anomaly and fraud detection, image denoising. Variational autoencoders (VAEs) can also generate new images.

10 Generative Adversarial Network (GAN)
GenAI · Vision

Two networks compete: a generator makes fakes and a discriminator tries to catch them. Both improve until the fakes look real.

Use cases: realistic faces, image upscaling, synthetic training data.

11 Diffusion Model
GenAI · Vision

Learns to remove noise step by step. To create, it starts from pure static and gradually "denoises" it into a new image that matches a text prompt.

Use cases: text-to-image tools, image editing and inpainting, video generation.

Learns the shape of the data without labels; text prompts are added as guidance.

HOW THEY COMPARE AT A GLANCE

ALGORITHM	LEARNING	TABULAR	VISION	NLP	GENERATIVE AI	INTERPRETABILITY
Linear / Logistic Regression	Supervised	●		○		High
Decision Tree	Supervised	●				High
Random Forest / Gradient Boosting	Supervised	●				Medium
K-Nearest Neighbors	Supervised	●	○			High
CNN	Supervised		●		○	Low
Transformer	Supervised*		○	●	●	Low
K-Means	Unsupervised	●	○			High
PCA	Unsupervised	●	○			Medium
Autoencoder	Unsupervised	○	●		○	Low
GAN	Unsupervised		○		●	Low
Diffusion Model	Unsupervised		○		●	Low

Solid dot = primary domain · faded dot = secondary use · *self-supervised pre-training, then supervised fine-tuning

FROM CLASSIC ML TO GENERATIVE AI

CLASSIC MACHINE LEARNING
Regression · Trees · KNN · K-Means · PCA
Works best on structured rows and columns. Fast, needs less data, and the results are easy to explain.

→

DEEP LEARNING
CNN · Autoencoder
Neural networks learn features on their own from raw pixels, audio or text. Needs a lot more data and computing power.

→

GENERATIVE AI
Transformer · GAN · Diffusion
Instead of only predicting a label, these models create new text, images and code by learning how the data itself is structured.

CHOOSING THE RIGHT ALGORITHM

Do you have labels?
Yes → supervised (predict a known answer). No → unsupervised (find groups, compress, or detect anomalies).

What does the data look like?
Spreadsheet-style → tree ensembles or regression. Images → CNNs. Text → Transformers.

Do you need to explain it?
In regulated areas like finance and security, a simpler model you can explain is often better than a slightly more accurate black box.

Sources: course readings on AI, ML, DL and GenAI; scikit-learn documentation